Week-1

Tuesday, May 20, 2025 6:03 AM

Week 1 (06/05/25)

- **Topics**: Foundations of regression, Simple Linear Model, Correlation, Estimation, Testing.
- **Homework**: HW#1 due 06/05/25.

Foundations and Ideas

- **Statistical Foundations**: Understanding probability, random variables, and distributions is crucial. Key concepts include:
  - **Descriptive Statistics**: Summarizing data using measures like mean, median, mode, variance, and standard deviation.
  - **Inferential Statistics**: Making predictions or inferences about a population based on sample data. This involves hypothesis testing and confidence i ntervals.
- **Theory of Estimation**: Focuses on methods for estimating unknown parameters using sample data. Key principles include:
  - **Bias and Consistency**: An estimator is unbiased if its expected value equals the true parameter value. Consistency means that as sample size incre ases, the estimator converges to the true parameter.

Simple Linear Model

- **Definition**: A statistical method that models the relationship between two variables by fitting a linear equation to observed data. The model is expressed as:
  Y=β0+β1X+ϵY = \beta_0 + \beta_1 X + \epsilonY=β0+β1X+ϵ
  where:
  - YYY = dependent variable
  - XXX = independent variable
  - β0\beta_0β0 = intercept
  - β1\beta_1β1 = slope
  - ϵ\epsilonϵ = error term
- **Assumptions**: The model assumes linearity, independence of errors, homoscedasticity (constant variance of errors), and normality of error terms.
- **Applications**: Used in various fields like economics, biology, and social sciences for predictive modeling and trend analysis.

Correlation

- **Definition**: A statistical measure that describes the strength and direction of a relationship between two variables. The most common measure is Pearson's correlation coefficient (rrr), which ranges from -1 to 1:
  - r=1r = 1r=1: perfect positive correlation
  - r=−1r = -1r=−1: perfect negative correlation
  - r=0r = 0r=0: no correlation
- **Types of Correlation**:
  - **Positive**: As one variable increases, the other also increases.
  - **Negative**: As one variable increases, the other decreases.
- **Limitations**: Correlation does not imply causation; two variables can be correlated without one causing the other.

Estimation

- **Point Estimation**: Provides a single value estimate of a parameter (e.g., sample mean as an estimate for population mean).
- **Interval Estimation**: Provides a range within which the parameter is expected to lie, often expressed as a confidence interval (e.g., 95% CI).
- **Methods**:
  - **Maximum Likelihood Estimation (MLE)**: Estimates parameters by maximizing the likelihood function.
  - **Method of Moments**: Estimates parameters by equating sample moments to theoretical moments.

Testing

- **Hypothesis Testing**: A method to determine if there is enough evidence in sample data to infer that a certain condition holds for the entire population.
- **Steps**:
  1. **State Hypotheses**: Null hypothesis (H0H_0H0) vs. alternative hypothesis (H1H_1H1).
  2. **Choose Significance Level (α\alphaα)**: Commonly set at 0.05.
  3. **Select Test**: Use appropriate statistical test (e.g., t-test, chi-square test).
  4. **Calculate Test Statistic**: Based on sample data.
  5. **Make Decision**: Compare the test statistic to critical values or p-values to accept or reject H0H_0H0.
- **Types of Tests**:
  - **One-sample t-test**: Compares the sample mean to a known value.
  - **Two-sample t-test**: Compares means from two different groups.
  - **ANOVA**: Compares means across three or more groups.